GROUND TRUTH UNDER ATTACK · TRACK 2 · FALSE DATA INJECTION

Project Specification

Cahier des charges · Phase 1

> We build a tool that tells if the numbers sent by a weather station can be trusted. It signs every reading, compares the station with independent witnesses (NASA POWER and Sentinel-2), gives the station a trust score, and stops false data before it changes a decision.

TEAM	OFFICIAL DEADLINE	BUDGET
4 members	**3 Nov 2026**	**0**
one clear role each	23:59 GMT+1 · we submit on 2 Nov	free software, one laptop, no hardware

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1 The project in short

A weather station stands alone in a field. Every day it sends temperature, humidity, wind, rain and sunshine. People use these numbers to decide about irrigation, drought help and crop insurance.

The problem

Someone can change the numbers to win something. For example, a station that reports less rain than the truth can trigger an insurance payment. The false numbers still look normal, so nobody notices.

The attacker can be at the station, on the network, or inside the storage system.

Our answer

We never trust one sensor alone. We do three things:

- **Sign** each reading, so it cannot be changed on the way.
- **Compare** the station with independent witnesses that the attacker cannot touch.
- **Score** the station from 0 to 1 and react step by step.

Two ideas that guide everything

Idea	What it means in simple words
Look for disagreement, not for extreme values	A heatwave is extreme but real: the station, NASA POWER and the satellite all agree. In an attack, the station disagrees with at least one witness.
Small proof adds up	The official attacks are small (12 % and 15 %). One day shows nothing. A small difference that stays day after day is not noise, so we add the evidence over days.

The context: the challenge

Item	Fact
Challenge	Ground Truth Under Attack: Securing the Earth Observation Chain
Our track	Track 2: False Data Injection on in-situ sensors (sensors on the ground)
Official data	NASA POWER Daily API (daily weather reference) and Copernicus Sentinel-2 L2A (satellite images)
Official test	A 12 % step bias, a slow drift up to 15 %, and a replay of 7 samples. The site and the dates are hidden in the final test.
Phase 1	Anonymous report + code. Worth 30 % of the final score. Up to 7 teams go to Phase 2.
Phase 2	Working prototype + live demo + pitch (7 min + 3 min questions). Worth 70 %. Deadline 11 December 2026.



How the final score is built. This document covers Phase 1.

2 Goals and scope

Goals

No.	Goal	How we know it is reached
G1	Detect the three official attacks at a place we have never seen.	Results shown on at least two sites that were never used to build or tune the tool.
G2	Do not confuse real bad weather with an attack.	False alarms are counted separately during real heatwaves, cold spells and heavy rain.
G3	Stop a false reading from changing a decision.	Same attack, run **without** and **with** our defence: the decision is wrong without, correct with.
G4	Stay cheap and simple.	Runs on a normal laptop, processor only, with free software. Nothing to buy.
G5	Anyone can repeat our results.	A team member who did not write the code gets the same main table on another laptop with one command.
G6	Submit on time and fully anonymous.	Submitted on 2 November. Anonymity checklist done.

Scope

Inside the project
One sensor type: a land weather station with daily values
Real public data; attacks are added by us on top of real records
The three official attacks first, then our own extra attacks
Signing and checking of readings, in software
Two detectors: one statistical, one learned
Trust score and four graded actions
One simple decision product: a 30-day water balance
Report, repository, tests, documentation

Outside the project (Phase 1)
Real sensor hardware or a real radio link
Deep learning and graphics cards (GPU)
Downloading full satellite images (we read small windows only)
A dashboard or web interface (kept for Phase 2)
Hiding the data (weather data is public, so secrecy is not needed)
A new farming model (we use a standard, documented one)
Sub-daily data (the official reference is daily)
Other tracks (ground link, image forgery)

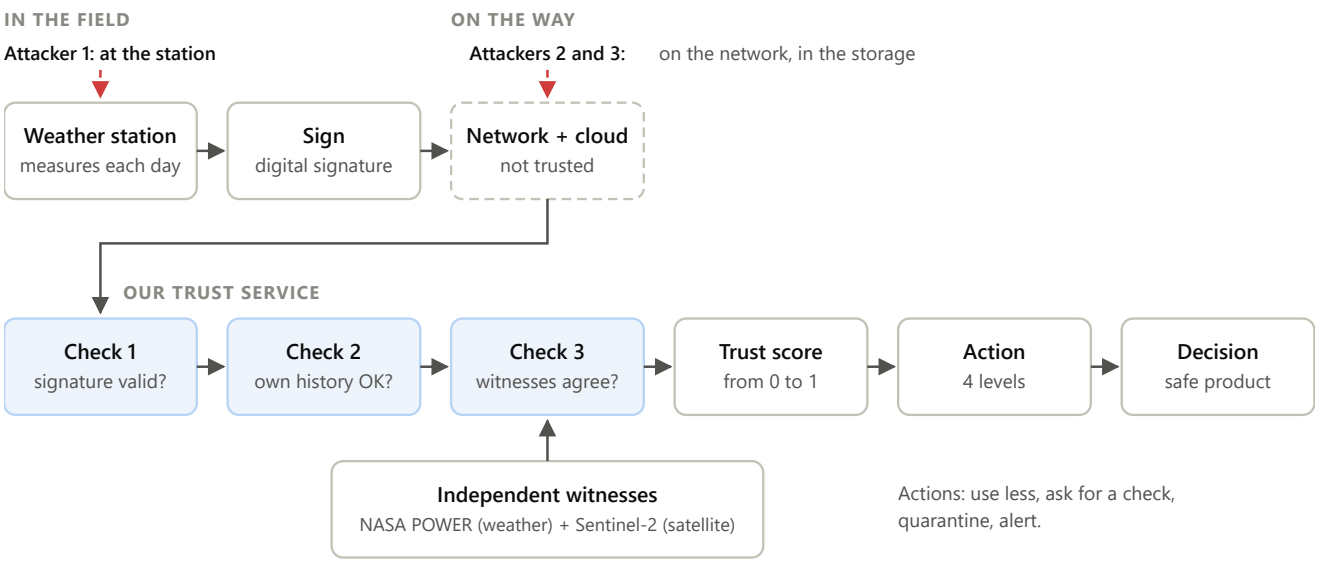
Who uses the result

Who	What they need from us
Jury	A clear report, numbers they can check, and a tool that works on their hidden test series.
Analyst (irrigation, drought, insurance)	A decision they can trust, and a clear warning when the data behind it is weak.
Station operator	An alert that says which station, since when, and why.
The team	Clear tasks, clear owners, and fixed data formats so four people can work at the same time.

3 How the system works

One reading travels from the station to the decision. On the way it passes three checks. Each check catches what the one before cannot see.

Figure 1. The path of one reading, and where an attacker can act.



The three checks

Check	Question	What it catches	What it cannot see
1 · Signature	Does this reading really come from this station, unchanged and new?	Fake, changed or replayed messages on the network or in the storage. Missing messages.	A station that is itself hacked or tricked: it signs false data with a valid key.
2 · Own history	Does the reading fit the station's own past, the season, and the physics between its values?	Frozen sensor, spikes, impossible values, a simple replay.	A small bias or a slow drift. They look like normal weather.
3 · Witnesses	Does the station agree with NASA POWER and Sentinel-2 for the same place and day?	A station that moves away from the witnesses, however the lie was made.	Changes smaller than the normal gap between a station and the reference.

From trust score to action

Trust level	State	What the system does
High	Trusted	The data is used normally.
Reduced	De-weighted	The data counts less. A **verification request** is sent to the operator.
Low	Quarantined	The data is blocked. Values are replaced by the calibrated reference. An **alert** gives the reason and the estimated start date.
No data	Silent	The station stopped sending. It is treated as unknown, never as "nothing happened".

Two ways to run the tool

Mode	Input	Used for
File mode	A daily series + latitude and longitude	The official test: the jury gives a series, the tool returns trust scores, detected attacks and actions.
Stream mode	Signed readings arriving one per day	The full demonstration, and the comparison with and without the defence.

4 Examples

> **Read this first.** All numbers in this section are **made-up examples** to explain the idea. They are not measured results. Real results will come from the tests in section 10.

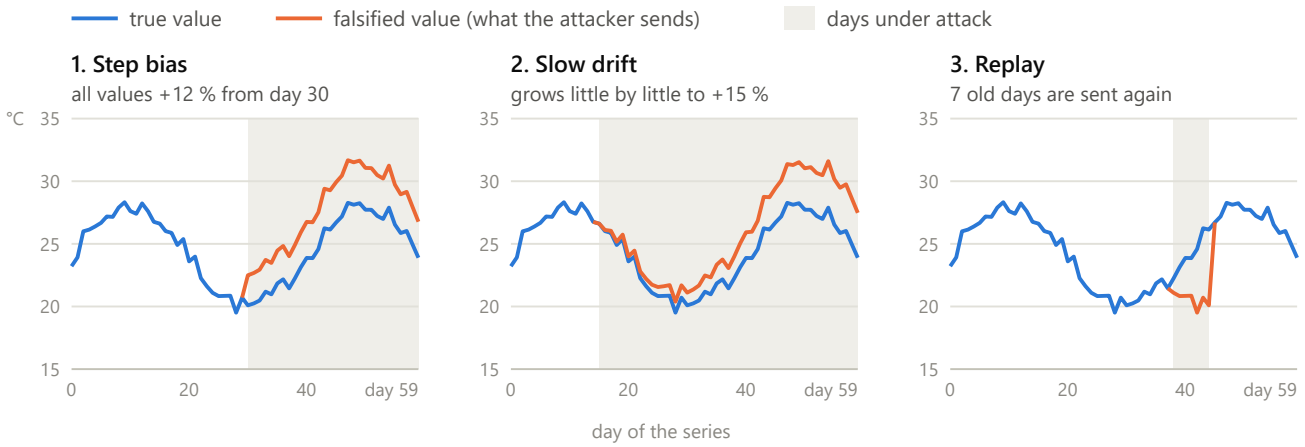
Example 1 · One signed reading

Field	Example value	Why it is there
Station	ST-0007	Says which station is speaking.
Counter	412	Goes up by one each day. An old number means a replay.
Date	14 July	The day of the measurement.
Values	31.4 °C · 38 % humidity · 3.2 m/s wind · 0.0 mm rain	The measurements of the day.
Link to yesterday	9f2c…a41b	A fingerprint of reading 411. A gap or a change in the past becomes visible.
Signature	64 bytes	Made with the station's secret key. Any change breaks it.

The security fields add 118 bytes to each reading (design value; the real cost will be measured).

Example 2 · The three official attacks

Figure 2. What each official attack does to a temperature series of 60 days.



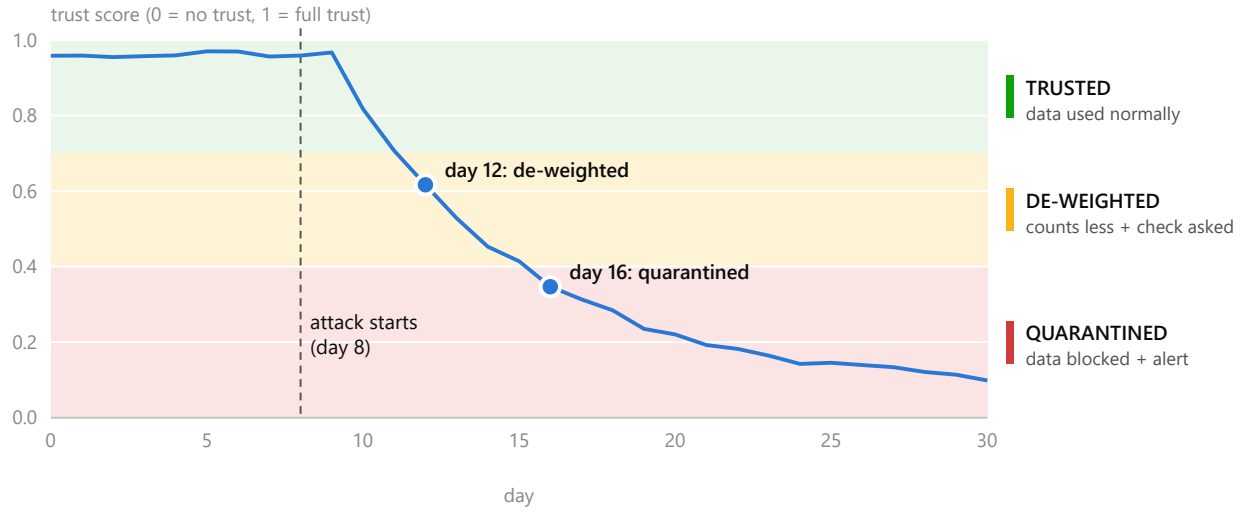
Made-up data for illustration. Notice that each falsified curve still looks like normal weather. That is why we need witnesses.

Example 3 · Four situations, four results

Situation	Check 1	Check 2	Check 3	Result
Normal day	pass	pass	pass	Trusted. Data used.
Real heatwave	pass	unusual	pass: witnesses also see the heat	Trusted. **No false alarm.**
Old message sent again on the network	**fail**: counter already seen	—	—	Rejected at once.
Hacked station adds 12 % and signs it	pass	pass	**fail**: the gap with the witnesses stays	Trust falls day by day, then quarantine and alert.

Example 4 · The trust score during an attack

Figure 3. Trust score of one station. A 12 % step attack starts on day 8.



Made-up data for illustration. The limits 0.7 and 0.4 are examples; the real limits will be set on validation data and then frozen.

The score does not fall on the first day, because one day proves nothing. It falls when the difference with the witnesses stays. In this example the station counts less from day 12 and is blocked from day 16.

Example 5 · What the tool gives back (file mode)

Detected problem	Estimated start	End	Lowest trust	Action
Step bias	day 30	still active	0.12	Quarantine + alert
Slow drift	day 18	still active	0.31	Quarantine + alert
Replay of 7 days	day 38	day 44	0.55	De-weight + verification request

Made-up output. The real tool also returns one trust score per day.

5 What the system must do

Each line is one function of the system. The number in the priority column is the **build order**. Orders 1 to 6 must be in the submission. Order 7 makes it stronger. Order 8 waits for Phase 2. If time runs short, we cut from the bottom.

ID	The system must…	Priority · order	Owner
F1	Download the daily weather reference (NASA POWER) for any place and dates. Keep a local copy with a checksum.	Must · 1	M1
F2	Load a real, public weather-station series (temperature, humidity, wind, rain, sun).	Must · 1	M1
F3	Inject the three official attacks into real data and keep exact labels (where the attack starts and ends).	Must · 1	M3
F4	Clean the data: remove impossible values, put station and reference on the same days, handle gaps.	Must · 2	M2
F5	Calibrate the station against the reference and compute the differences (called *residuals*).	Must · 2	M2
F6	Statistical detector: add up small differences over days, find change points, find repeated days.	Must · 2	M4
F7	File mode: one command takes a series + coordinates and returns trust scores, detected attacks and actions.	Must · 2	M2
F8	Measure the five required metrics and write the result tables automatically.	Must · 2	M4
F9	Sign every reading at the station (key per station, counter, link to the previous reading).	Must · 3	M3
F10	Verify every reading: reject fake, changed or replayed readings; notice missing ones.	Must · 3	M3
F11	Give each reading a compromise probability and each station a trust score between 0 and 1.	Must · 3	M4
F12	Apply one of four actions from the trust score: use less, ask for a check, quarantine, alert.	Must · 3	M3
F13	Build the decision product (30-day water balance) and compare **with** and **without** the defence.	Must · 3	M2

ID	The system must…	Priority · order	Owner
F14	Keep a log that cannot be changed silently (each entry is chained to the one before).	Must · 3	M3
F15	Read Sentinel-2 indicators around the site (cloud, snow, vegetation, moisture) and compare.	Must · 4	M1
F16	Learned detector (trained on clean data only), compared with the statistical one on the same data.	Must · 5	M4
F17	Evasion tests: a smart attacker tries to hide. At least one test per defence layer.	Must · 6	M3 M4
F18	One command reproduces every table and figure of the report.	Must · 6	M2
F19	Extra attacks beyond the official three (frozen sensor, spikes, missing data, date shift…).	Good to have · 7	M3
F20	Operator screen (dashboard) showing station trust live.	Phase 2 · 8	—

M1 to M4 are the four team members. Their roles are in section 8.

6 Quality rules and targets

Rules the whole project must respect

ID	Rule	How we check it
Q1	Runs on a normal laptop, processor only. No GPU, no hardware to buy.	Full run on two different laptops.
Q2	One command reproduces everything.	Fresh copy of the repository, one command, same main table.
Q3	Works for any place on Earth. Nothing depends on one station.	Tests on sites never seen before; input is only a series + coordinates.
Q4	Works without internet after the first download.	Run again with the network off, using the local copy.
Q5	Fully anonymous.	Anonymity checklist (section 12) before submission.
Q6	Honest test: the test data is never used for building or tuning.	Data split by time and by site. Limits frozen before the test.
Q7	Same result every time.	Fixed random seeds, fixed library versions.
Q8	Every main part has automatic tests.	All tests pass on a fresh copy.
Q9	No secret in the repository (accounts, keys, tokens).	Review before the export.

The five numbers the jury wants

Metric	Simple meaning	What we report
Detection quality	How many attacks do we catch?	Share of attacks detected, for each attack and each measured value.
False alarms	How often do we cry wolf?	False alarms per station per year, and separately during real extreme weather.
Latency	How fast do we react?	Days between the start of the attack and the first flag, and the quarantine.
Memory	How much memory do we use?	Highest memory used, model size on disk.
Security overhead	What does security cost?	Extra bytes per reading, time to sign and to verify.

Proposed targets

Topic	Proposed target
Step bias 12 %	Detected in 14 days or less on temperature, humidity and solar radiation.
Drift up to 15 %	Detected before it reaches its full size.
Replay of 7 samples	Detected before the 7 days are over.
False alarms	2 or fewer per station per year; no quarantine caused by real extreme weather.
Run time	Quick mode in 5 minutes or less; full run in 30 minutes or less on a laptop.
Wrong decisions	With the defence, clearly fewer wrong decisions than without, for each official attack.

> **These targets are a first proposal, not a promise.** No code exists yet, so nobody has measured them. The team confirms or changes them at gate G2 (19 October), when the first real numbers exist. We already expect that a 12 % change will be hard to see on wind and on daily rain; the report will say so openly.

7 What we hand in

Four things to submit

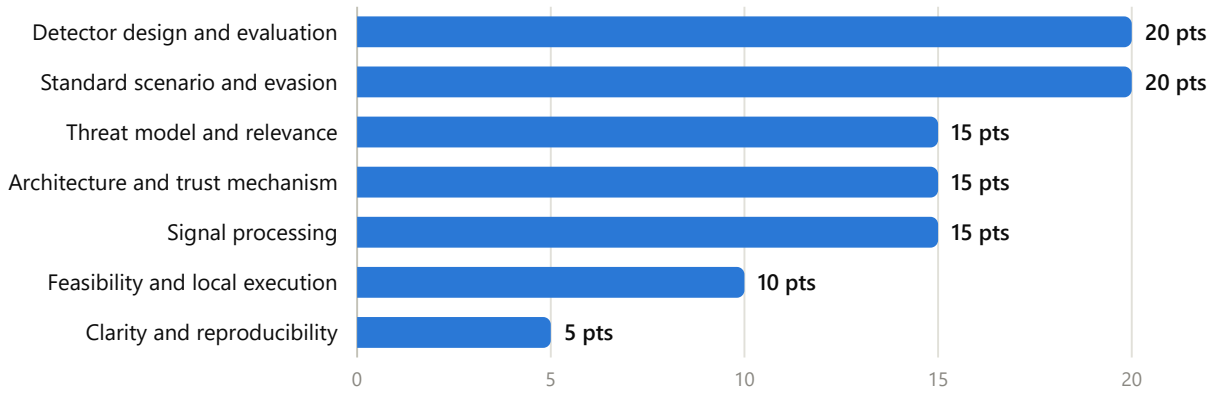
Item	Content	Finished when
Anonymous report	PDF that covers the six parts below, the results and the limits.	Each scoring line has a section with at least one figure or table. PDF has no author data.
GitHub repository	Code, test scenarios, tests, documents, fixed library versions.	A fresh copy reproduces every number of the report with one command.
Prototype	The full chain: reading, sign, send, attack, verify, compare, detect, score, act, decide.	The full run ends on a laptop and writes the tables and figures.
Documentation	README, the docs folder, how to get the data.	A stranger can install, run and understand it without asking us.

Six parts the report must contain

Part	In simple words	Lead
D1 · Threat model	Who can attack, where, how, and what it costs them.	M3
D2 · Architecture and trust anchor	The system diagram, the keys, the signed reading, how we verify.	M3
D3 · Signal processing	Cleaning, alignment, calibration, and the differences with the witnesses.	M2
D4 · Detection	Two detectors compared, the probability and the trust score.	M4
D5 · Attack tests and response	Official test, blind rehearsal, evasion tests, with / without, the four actions.	M4 M3
D6 · Documentation with the official data	Dataset table, download scripts, one-command run, hardware and run time.	M1

Where the points are

Figure 4. Phase 1 scoring: 100 points in seven lines.



40 of the 100 points are about testing (the two top bars). A simple design with careful numbers beats a beautiful design with few numbers.

8 The team of four

The plan has five work areas. With four members, the fifth area (integration and report) is shared. Each member has one main area and one extra job. Names are not written here because Phase 1 is anonymous.

Member	Main area	Owns (code folders)	Extra job	Reviewed by
M1	**Data and remote sensing**	`data` : NASA POWER, Sentinel-2, station loaders, site choice	Report editor: assembles and polishes the report	M2
M2	**Signal processing**	`processing` , `product` : cleaning, calibration, differences, decision product	Integration: one-command run, repository, README	M1
M3	**Security and attacks**	`trust` , `sensor` , `transport` , `attacks` , `response`	Threat model text; anonymity check and clean export	M4
M4	**Detection and evaluation**	`detect` , `eval` : detectors, trust score, metrics	All result tables and figures	M3

Who does what

L = leads (does the work and answers for it) · **H** = helps (does a part) · **R** = reviews before it is merged · empty = kept informed

Work	M1	M2	M3	M4
Reference data, sites, dataset table	**L**	**R**		**H**
Cleaning, calibration, differences	**R**	**L**		**H**
Signing, verification, log		**H**	**L**	**R**
Attack and evasion library			**L**	**R**
Threat model	**H**		**L**	**R**
Detectors and trust score		**H**	**R**	**L**
Metrics, tables, figures	**H**		**R**	**L**
Response policy (four actions)		**H**	**L**	**R**
Decision product, with / without	**R**	**L**		**H**
One-command run, repository, README	**R**	**L**	**H**	**H**
Blind rehearsal (hidden attack)	**H**		**L**	**H**
Report: own sections	**L**	**L**	**L**	**L**
Report: assembly and final edit	**L**	**R**	**H**	**H**
Anonymity check and clean export	**R**	**H**	**L**	

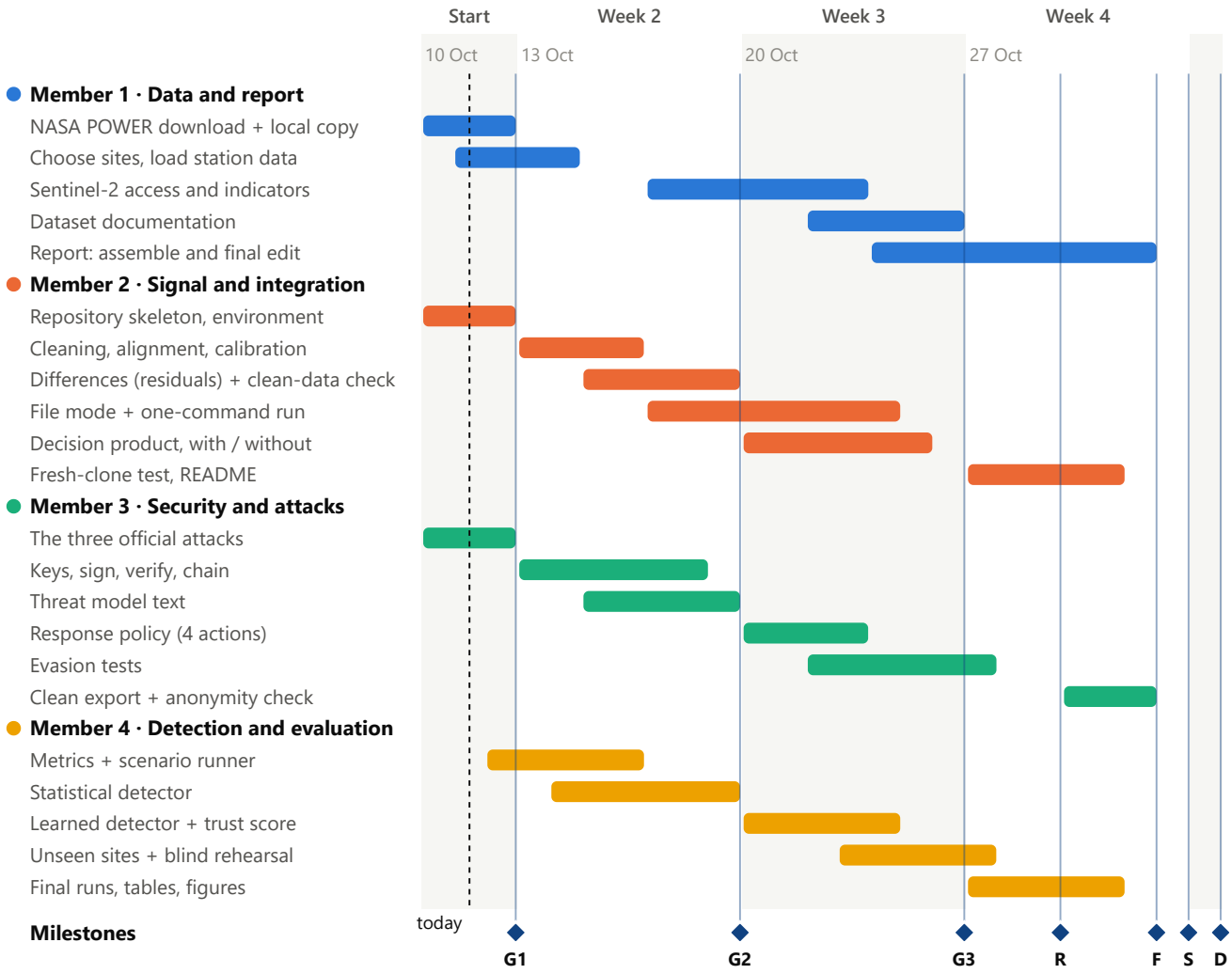
Working rules

- **Agree the shared formats first** (by 13 October): the table of readings, the list of differences, the attack labels, the settings file. After that, four people can work in parallel.
- **Short check-in every day** (15 minutes): what I did, what I do next, what blocks me.
- **Gate meeting on each gate day** (12, 19, 26 October): we look at the result together and decide what to cut.
- **One branch per task**, reviewed by the partner in the table above before merging.
- **Write the report as we go**: each section is drafted in the week its work is done.
- **Backup pairs**: M1 with M2, M3 with M4. If one member is away, the partner can continue.

9 Schedule

24 days from 10 October to the planned submission on 2 November. One day is kept as a safety margin before the official deadline.

Figure 5. Who works on what, and when (10 October to 3 November 2026).



Colours show the member: M1 M2 M3 M4. Diamonds are milestones, explained below.

Milestones

Code	Date	Milestone	It is reached when…
G1	Mon 12 Oct	Data and attacks ready	For 3 sites: a plot of station against reference on clean data, and the three attacked series made by one command.
G2	Mon 19 Oct	First numbers	The official test runs with one command and prints detection, latency and false alarms. The differences stay small in real extreme weather.
G3	Mon 26 Oct	All tables filled	Every result table of the report has numbers, even if rough.
R	Thu 29 Oct	Report complete	Full draft ready. Internal review on Friday 30.
F	Sun 1 Nov	Freeze	No more changes to code or report.
S	Mon 2 Nov	Submission	Report and clean repository sent.
D	Tue 3 Nov	Official deadline	23:59 GMT+1. Safety day only.

After Phase 1: preselection results on 8 November, final deadline on 11 December 2026.

10 Tests and acceptance

The official test

Attack	Official size	How we build it (until the organizers say more)
Step bias	12 %	Every value is multiplied by 1.12 or 0.88 from a random day.
Slow drift	up to 15 %	A straight ramp from 0 to 15 % over 30, 60 and 90 days.
Replay	7 samples	7 days in a row are replaced by 7 earlier days of the same series.

Other tests

Test	What we do	Why
Clean data	Run on real data with no attack, including real extreme weather.	Counts the false alarms.
Unseen sites	Run on at least two sites never used before.	Copies the hidden site of the final test.
Blind rehearsal	One member hides an attack at a place and dates the others do not know. The others run the tool once.	An honest test of ourselves. The result is reported as it is.
With / without	Same attack, with no defence and with the full defence.	Shows that the defence protects the decision.
Layer by layer	Compare: nothing · basic quality control · signature only · + own history · full system.	Shows that each layer is useful.
Evasion	A smart attacker goes slower, stays in normal ranges, or copies sensor ageing.	Finds where the defence stops working. We report it openly.

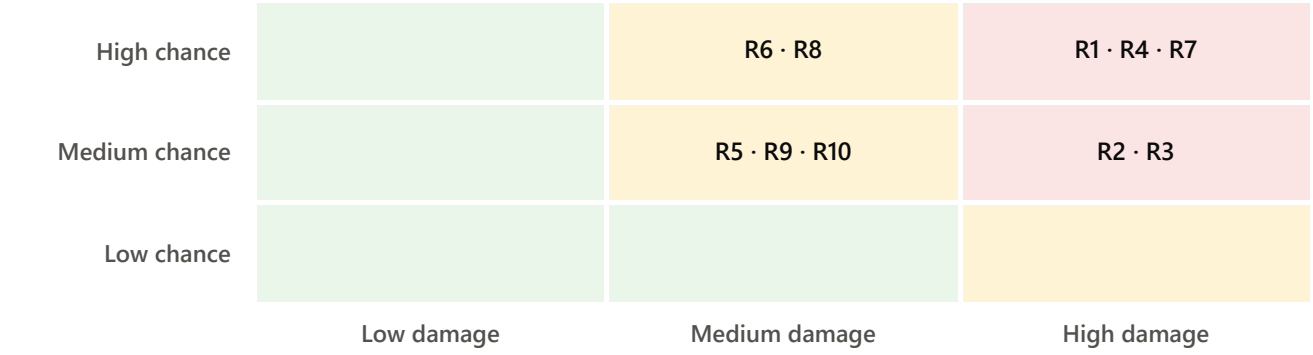
Phase 1 is done when...

- [ ] The report covers the six parts and the five metrics.
- [ ] The official test is reported on the official data, on known and unseen sites, for each measured value.
- [ ] One command reproduces every table and figure on a laptop.
- [ ] File mode takes a series and coordinates and returns trust scores, detected attacks and actions.
- [ ] The with / without comparison shows the decision changing without the defence and not with it.
- [ ] False alarms are reported separately for real extreme weather.
- [ ] There is at least one evasion test for each of the three checks.
- [ ] All automatic tests pass on a fresh copy.
- [ ] The anonymity checklist is complete, with the export to a clean repository.
- [ ] Everything is submitted on 2 November.

11 Risks

ID	Risk	Chance	Damage	What we do about it	Owner
R1	A 12 % change is too small to see on some values (wind, daily rain).	High	High	Measure it in week 2. Report results per value. Ask the organizers which values are tested.	M4
R2	The organizers mean something else by “12 %” or attack another series.	Medium	High	Ask now. Keep every meaning runnable from a settings file.	M3
R3	The detector fails on the hidden site.	Medium	High	Nothing tuned to one site. Unseen sites and blind rehearsals.	M4
R4	Real bad weather is flagged as an attack.	High	High	Detect on differences, not on raw values. Clean-data check at gate G2.	M2
R5	Sentinel-2 is hard to get (account, formats, clouds).	Medium	Medium	Register in the first days. The system also runs without it.	M1
R6	Too many extras (dashboard, deep learning, more sensors).	High	Medium	Follow the build order of section 5. Cut from the bottom.	All
R7	Names leak through the git history or file data.	High	High	Export to a new repository with a clean history. Checklist.	M3
R8	The report is written in the last two days.	High	Medium	Each section is drafted in the week its work is done.	M1
R9	Results differ on another laptop.	Medium	Medium	Fixed versions and seeds. Fresh-copy test in week 4.	M2
R10	A member is not available for some days.	Medium	Medium	Backup pairs. Shared formats written down. Small tasks, merged often.	All

Figure 6. Risk map: where each risk sits.



Red cells = handle first. R1, R4 and R7 need action in the first two weeks. R7 is certain to happen if we do nothing: the current git history already contains a real name.

12 Tools, budget, rules

Tools

Need	Tool	Cost
Language	Python 3.11 or newer	Free
Tables and maths	pandas, NumPy, SciPy	Free
Change points	`ruptures` (listed by the organizers)	Free
Anomaly detection	PyOD, scikit-learn (PyOD is listed by the organizers)	Free
Signatures	`cryptography` or PyNaCl (Ed25519, SHA-256)	Free
Figures	matplotlib	Free
Tests	pytest	Free
Experiments	YAML settings files: one file per experiment	Free
Data	NASA POWER (no account) · Sentinel-2 through Copernicus Data Space (free account)	Free
Hardware	Our own laptops. No sensor, no radio, no server.	0

Total budget: 0. The challenge gives no advantage for bought hardware.

Anonymity rules (Phase 1)

- [ ] No names, university, student branch, or logos in the report, README, code, figures or licence file.
- [ ] The submission is exported to a **new repository with a clean history** and a neutral git name and e-mail.
- [ ] PDF file data is cleared (author, creator, company).
- [ ] Notebook outputs are cleared. No file path with a user name.
- [ ] No screenshot showing a user name, a desktop or a chat.
- [ ] No account name, key or token in the repository.

Open questions for the organizers

No.	Question	Why it matters
1	Which series is attacked: a station series, or a NASA POWER series? Is it given as a file?	Changes how the official test is run.
2	Are 12 % and 15 % measured on the value, on its range, or on its spread? Which values?	Changes the size of the attack.
3	How long does the drift take? Where do the 7 replayed samples come from?	Changes the test settings.
4	Is a clean history given for the hidden site? Is internet available during the final test?	Decides if we must run offline.
5	How is the test scored, and how is a false alarm counted?	Decides which number we optimise.
6	Report format: page limit, template, file type? How do we submit the repository anonymously?	Needed before writing.

Until we get answers, we use the settings of section 10 and keep the other options ready in settings files.

13 Simple dictionary

Word	Simple meaning
In-situ sensor	A sensor that is physically at the place it measures (here: a weather station in a field).
False data injection	Putting wrong numbers into a data stream on purpose.
Step bias	From one day on, every value is too high or too low by the same share.
Drift	The error grows slowly, a little more each day.
Replay	Old real values are sent again as if they were new.
Digital signature	A code made with a secret key. It proves who sent the data and that nobody changed it.
Trust anchor	The first key that everybody trusts. All other keys are approved by it.
Provenance	The history of a piece of data: where it comes from, when, and what was done to it.
Reference data	Independent data for the same place and day, used as a witness (NASA POWER, Sentinel-2).
Calibration	Learning the normal gap between a station and the reference, so we can remove it.
Residual	The difference between what the station says and what we expect.
Change point	The day when the behaviour of a series changes.
Compromise probability	How likely it is that one reading is false (0 to 1).
Trust score	How much we trust a station (0 to 1). It falls fast and comes back slowly.
De-weighting	The data still counts, but less.
Quarantine	The data is set aside and not used until someone checks the station.
False alarm	The system says “attack” but the data was real.
Latency	The time between the start of an attack and its detection.
Overhead	The extra cost of security (bytes, time, memory).
Evasion	When an attacker changes the attack to avoid being detected.
Unseen site (held-out)	A place never used to build or tune the tool. Used only for the final check.
Gate	A fixed day when the team checks a result before going on.